Understanding Bluetooth Internet Gateways For IoT Solutions

Based on some work carried out by the Bluetooth SIG relating to the integration of Bluetooth devices with the Internet and IoT architecture, here are some interesting results

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According to the research organisation ABI, Bluetooth technology is in 38% of all IoT devices while Wi-Fi and cellular are in 32% and 19%, respectively. As they have different strengths and weaknesses, these are not necessarily mutually exclusive technologies and are often used together in IoT solutions.

But there is a problem! Bluetooth and TCP/IP are not compatible. (IoT implies the use of protocol like TCP/IP.)

As it can be seen, TCP/IP spans only two layers of the Bluetooth stack. Other protocols are also used within the stack TCP/IP and Friends (Fig. 1). However, Bluetooth LE cannot talk directly with TCP/IP and vice-versa.

Wide standards range

Although some standards that relate to IoT and Bluetooth have been defined by the Bluetooth SIG and others, it is unlikely that all aspects of IoT can be covered using them alone. Currently, the following standards are defined:

RFC7668. An Internet Engineering Task Force (IETF) standard for communication, which defines how IPv6 6LoWPAN packets can be adapted to be sent over the lower layers of the Bluetooth LE stack. Not all aspects of TCP/IP are supported but Bluetooth LE devices using this can have IP addresses and can communicate with each other using 6LoWPAN IPv6 packets.

Internet Protocol Support Profile (IPSP). This is a Bluetooth Low Energy (LE) profile that allows devices that support IP over Bluetooth to be discovered. It defines how the Bluetooth LE stack is used to send and receive IPv6 packets in RFC7668.

HTTP Proxy Service (HPS). This is useful for devices like sensors and allows a Bluetooth sensor or similar to work with a gateway device, which supports both

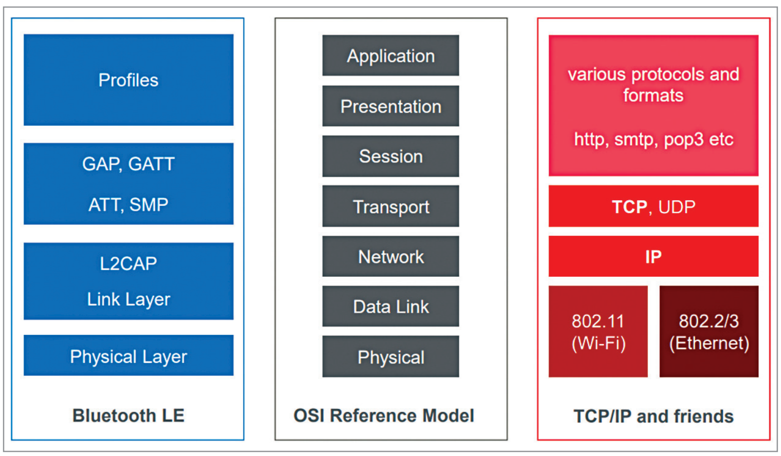

Fig. 1: Bluetooth LE vs TCP/IP

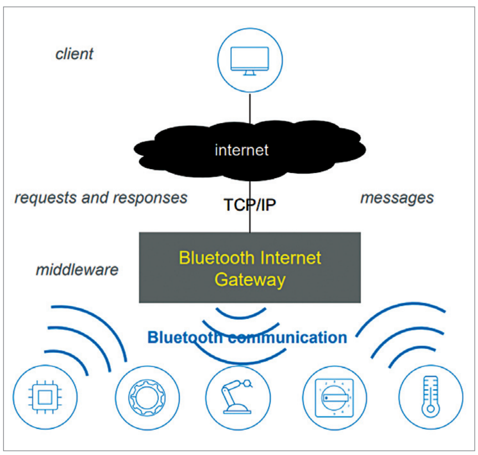

Fig. 2: Bluetooth and IoT solution architectures